

Hotel Booking Data Analysis – Portfolio Report

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Role: Data Analysis Project

Tools: Python, Pandas, Matplotlib, Seaborn

This portfolio report presents an exploratory data analysis of hotel booking demand. The objective is to uncover patterns in reservations, cancellations, and pricing to support better business decisions for hotel revenue management.

1. Project Overview

The hospitality industry relies heavily on accurate demand forecasting and pricing strategies. This project analyzes booking behavior across city and resort hotels.

The dataset contains booking records from 2015 to 2017. The analysis focuses on cancellation behavior, seasonal price fluctuations, and revenue opportunities.

Understanding these patterns allows hotel managers to optimize pricing, improve occupancy rates, and minimize revenue loss caused by cancellations.

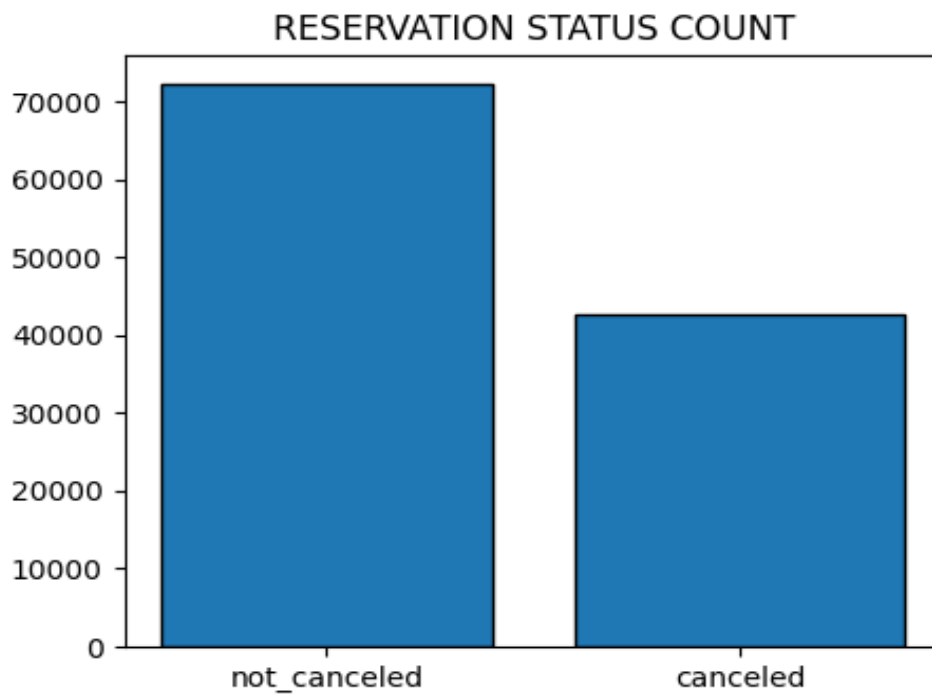
2. Dataset Description

Feature	Description
Dataset Size	~119,000 hotel bookings
Time Period	2015–2017
Hotel Types	City Hotel, Resort Hotel
Key Metric	ADR (Average Daily Rate)
Objective	Analyze cancellations and pricing trends

The dataset includes reservation information such as lead time, number of guests, market segment, hotel type, and booking status.

Average Daily Rate (ADR) is used as the primary revenue indicator in this analysis.

3. Reservation Status Analysis

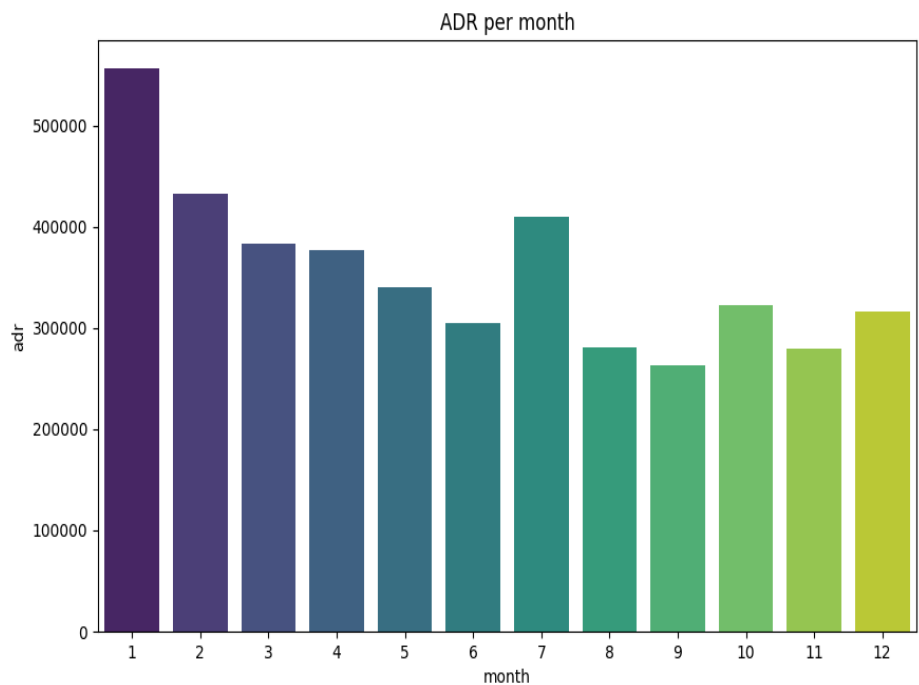


The reservation distribution indicates that approximately 72,000 bookings were completed while about 43,000 bookings were cancelled.

This results in a cancellation rate close to 37%, which is relatively high for the hospitality industry.

High cancellation rates create uncertainty in room availability and negatively impact revenue management.

4. Monthly Average Daily Rate (ADR)



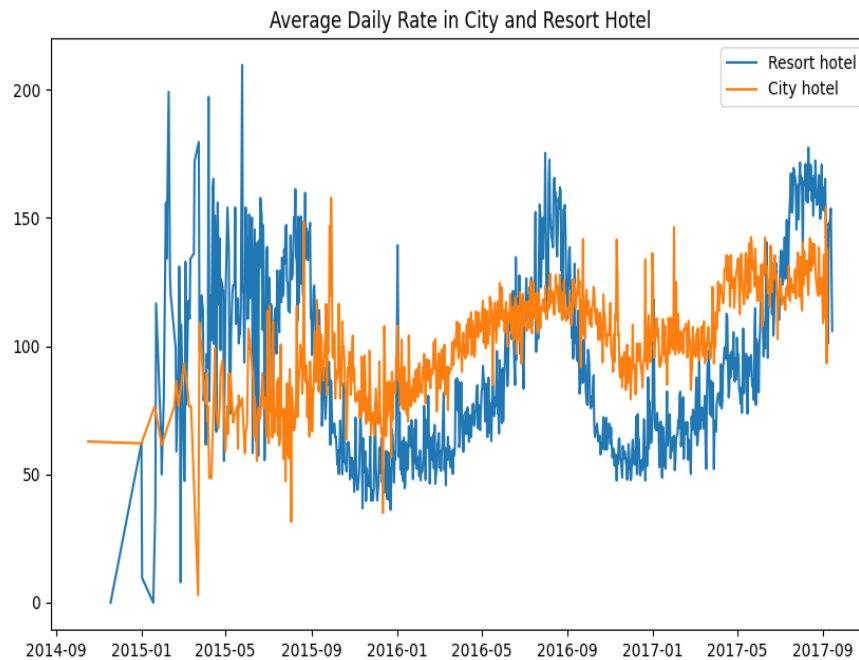
Month	ADR
Jan	560
Feb	430
Mar	380
Apr	375
May	340
Jun	305
Jul	410
Aug	280
Sep	265
Oct	320
Nov	280
Dec	315

The monthly ADR chart highlights strong seasonal trends in hotel pricing.

January shows the highest average price levels, while September records the lowest.

Seasonal demand strongly influences pricing strategies within the hotel industry.

5. ADR Comparison by Hotel Type

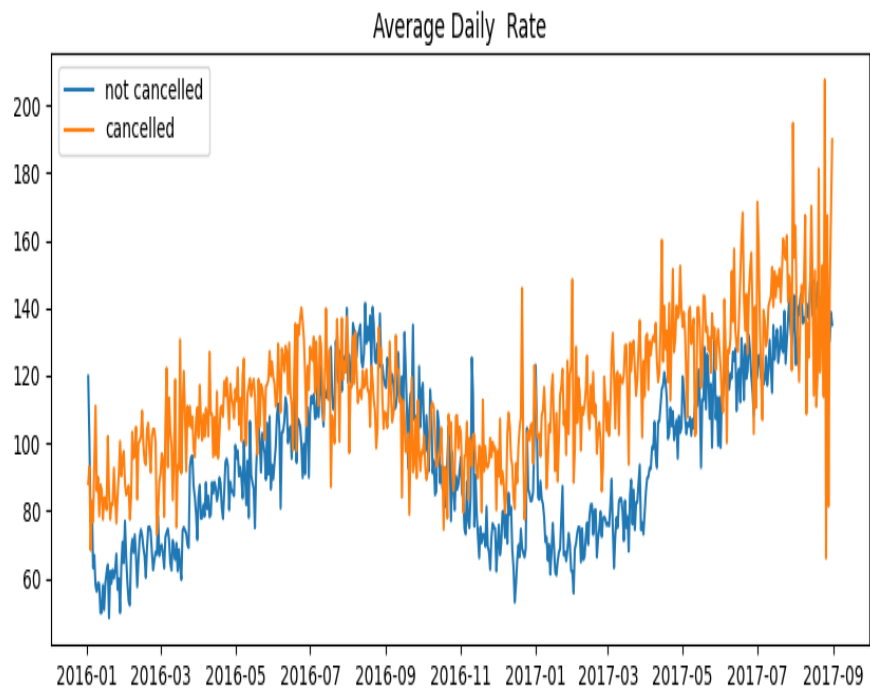


Resort hotels demonstrate significant volatility in ADR values.

This fluctuation occurs because resort demand depends heavily on tourism seasons.

City hotels display more stable pricing patterns due to consistent business travel.

6. ADR vs Reservation Status



The relationship between ADR and reservation status shows that higher priced bookings tend to cancel more frequently.

This suggests that customers booking expensive rooms may reconsider their decisions or find better alternatives.

7. Key Business Insights

1. Cancellation rate is relatively high and directly affects revenue forecasting.
2. Hotel pricing is strongly influenced by seasonal demand patterns.
3. Resort hotels rely heavily on tourism demand.
4. Higher ADR bookings show higher cancellation probability.

8. Strategic Recommendations

Implement partial payment policies to reduce cancellation risk.

Introduce dynamic pricing models based on demand forecasting.

Promote discount packages during low demand months.

Offer loyalty incentives for customers who complete their stays.

9. Key Performance Indicators

Metric	Value
Total Bookings	~115,000
Completed Bookings	~72,000
Cancelled Bookings	~43,000
Cancellation Rate	37%
Highest ADR	560 (January)
Lowest ADR	265 (September)

10. Conclusion

This analysis demonstrates the importance of data-driven decision making in the hospitality industry.

By understanding booking behavior and pricing patterns, hotels can improve revenue management and operational planning.

Future work may include predictive models to forecast cancellations and optimize pricing strategies.